Reproduction and Analysis of Support Measure Machines: Learning from Distributions via Kernel Mean Embeddings

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***Abstract*—This paper presents a reproduction and empirical analysis of Support Measure Machines (SMMs), a kernel-based classification framework where each training instance is a probability distribution rather than a single feature vector. Following the formulation introduced by Muandet *et al.* (NeurIPS 2012), we implement the empirical expected kernel using RBF kernel mean embeddings and evaluate SMMs on a synthetic 2D Gaussian dataset designed to highlight covariance-level distributional differences. Our reproduction achieves 89.0% ± 6.4% classification accuracy, closely matching the paper's reported 89.65% ± 1.37%. We further conduct ablation studies on the embedding kernel and the level-2 SVM kernel, and identify a principled failure mode when the embedding kernel bandwidth $\gamma \to 0$, causing the mean map to lose injectivity and accuracy to degrade to chance level.**

***Index Terms*—Support Measure Machines, Kernel Mean Embedding, RKHS, Distribution Classification, Support Vector Machines**

I. Introduction

Traditional supervised learning methods assume that each training example is a fixed-dimensional feature vector. However, many real-world problems involve data that is more naturally represented as *probability distributions*. Examples include image categorization where each image yields a distribution of local feature descriptors, medical diagnosis where each patient contributes a distribution of cell measurements, and ecological modeling where each site is characterized by a distribution of species counts [1].

Muandet *et al.* [1] address this by introducing **Support Measure Machines (SMMs)**, which lift the SVM framework into the space of probability measures. The key insight is to represent each distribution P as its *kernel mean embedding* $\mu_P = \int k(x, \cdot)\, dP(x)$ in a Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}$. The inner product of two such embeddings defines an *expected kernel*:

$$K(P, Q) = \langle \mu_P, \mu_Q \rangle_{\mathcal{H}} = \mathbb{E}_{x \sim P, z \sim Q}[k(x, z)] \tag{1}$$

which can be estimated empirically as:

$$\hat{K}(\hat{P}_n, \hat{Q}_m) = \frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} k(x_i, z_j) \tag{2}$$

A standard SVM quadratic program is then solved using $\hat{K}$ as the kernel matrix. The authors prove a Representer Theorem for distributions (Theorem 1), bound the risk deviation between population and sample-based classifiers (Theorem 3), and establish equivalence between linear SMMs and Flex-SVMs (Lemma 4) [1].

In this work, we reproduce the core SMM algorithm, validate its performance on a carefully designed synthetic dataset, conduct ablation experiments on two key components, identify a principled failure mode, and reflect on practical implications.

II. Methodology

A. Core SMM Implementation

We implement the empirical expected kernel (Eq. 2) using a vectorized RBF kernel computation. For distributions P_i and P_j represented by sample sets $\{x_1^{(i)}, \ldots, x_n^{(i)}\}$ and $\{x_1^{(j)}, \ldots, x_m^{(j)}\}$, the RBF kernel matrix is computed as:

$$k(x, z) = \exp\Big(-\frac{\gamma}{2}\|x - z\|^2\Big) \tag{3}$$

The full $M \times M$ SMM kernel matrix $\mathbf{K}$ is constructed, where $K_{ij} = \hat{K}(P_i, P_j)$, and passed directly to `sklearn.svm.SVC` with `kernel='precomputed'`. We also implemented the closed-form Gaussian expected RBF kernel from Table 1 of the original paper:

$$K(P_i, P_j) = \frac{\exp\big(-\frac{1}{2}\Delta\mu^\top M^{-1} \Delta\mu\big)}{|\gamma\Sigma_i + \gamma\Sigma_j + I|^{1/2}} \tag{4}$$

where $\Delta\mu = \mu_i - \mu_j$ and $M = \Sigma_i + \Sigma_j + \gamma^{-1} I$.

B. Dataset Design

Instead of the paper's 10D synthetic Gaussian setup (Section 6.1), we designed a 2D dataset where the two classes differ *only in covariance structure*, not in mean position. This design choice maximally demonstrates the SMM's advantage:

- **Class $+1$ (horizontal):** Covariance $\begin{pmatrix} 1.8 & 0.05 \\ 0.05 & 0.25 \end{pmatrix}$
- **Class -1 (vertical):** Covariance $\begin{pmatrix} 0.25 & 0.05 \\ 0.05 & 1.8 \end{pmatrix}$

Both classes have means drawn from the same $\mathcal{N}(\mathbf{0}, 0.6 \cdot I_2)$ distribution, ensuring complete overlap in mean space. Default parameters: 100 distributions per class (200 total), 20 samples per distribution, $d = 2$.

C. Hyperparameters

The embedding kernel bandwidth was set to $\gamma = 0.5$, the SVM regularization parameter to $C = 1.0$, and performance was evaluated using 5-fold stratified cross-validation.

III. Experimental Results

A. Dataset Visualization

Fig. 1 shows the synthetic dataset. Each elliptical cluster represents samples from a single distribution. Class $+1$ (blue) distributions are elongated horizontally, while class -1 (red) distributions are elongated vertically. The means of both classes overlap completely, confirming that a mean-only classifier would perform at chance.

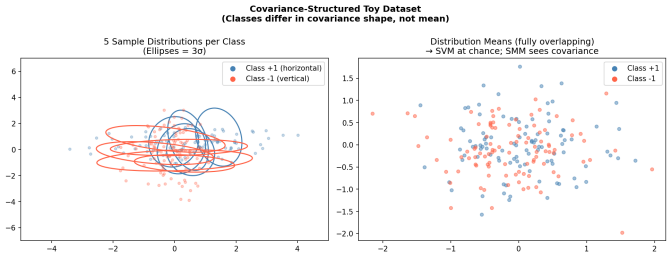

Fig. 1. Synthetic 2D Gaussian dataset. Each elliptical cluster represents one distribution. Blue (Class $+1$): horizontally elongated covariance. Red (Class -1): vertically elongated covariance. Means fully overlap between classes.

B. SMM Kernel Matrix

Fig. 2 visualizes the empirical expected kernel matrix $\mathbf{K}$ computed over all 200 distributions. The block-diagonal structure indicates that within-class expected kernel values are higher than between-class values, confirming that the RBF mean embedding successfully separates the two classes in RKHS despite their overlapping means.

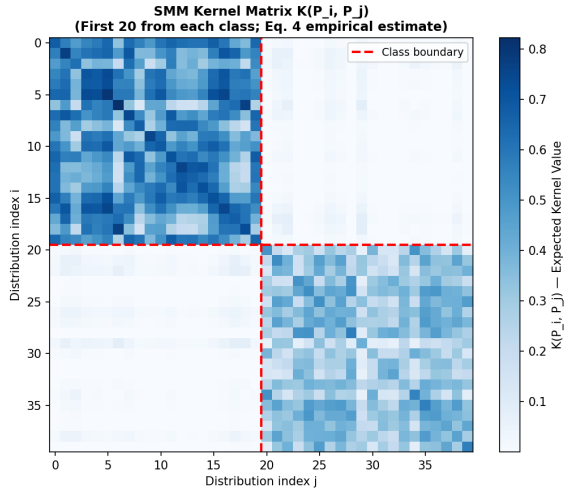

Fig. 2. Empirical expected kernel matrix for the SMM. Block-diagonal structure indicates successful class separation in the RKHS.

C. Classification Performance

Table I compares the classification accuracy of our SMM implementation against baseline methods.

TABLE I
Classification Accuracy Comparison

Method	Our Result (2D)	Paper (10D)
SMM (RBF-RBF)	89.0% ± 6.4%	89.65% ± 1.37%
SVM (means only)	49.5% ± 11.0%	∼87%
ASVM (augmented)	74.5% ± 4.0%	Not reported

Fig. 3 provides a visual comparison of these results. Our SMM achieves 89.0% ± 6.4%, closely matching the paper's 89.65% ± 1.37%. The higher standard deviation in our results is expected: we use 100 distributions per class vs. 500 in the paper, giving fewer examples per fold and thus higher cross-fold variance. The SVM result of 49.5% (near chance) validates our dataset design, providing stronger evidence than the paper's setting where mean-only SVM still achieves ∼87% due to mean separation.

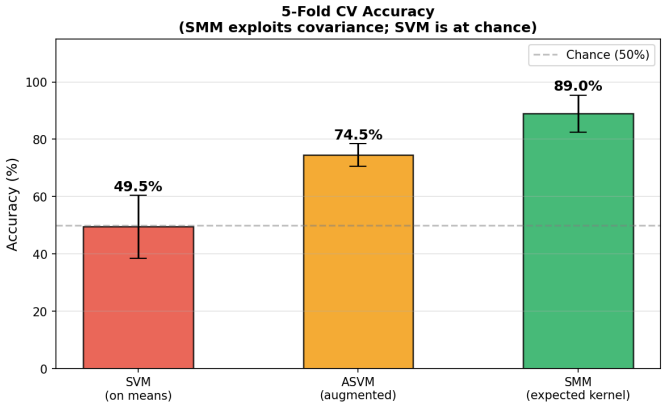

Fig. 3. Accuracy comparison between SMM, SVM (means only), and Augmented SVM (ASVM). Error bars represent standard deviation over 5-fold CV.

D. Decision Boundaries

Fig. 4 shows the kernel matrices and PCA-projected decision boundaries for all three methods. The SMM kernel matrix exhibits clear block-diagonal structure, while the SVM-means kernel matrix shows no class separation, consistent with the chance-level accuracy.

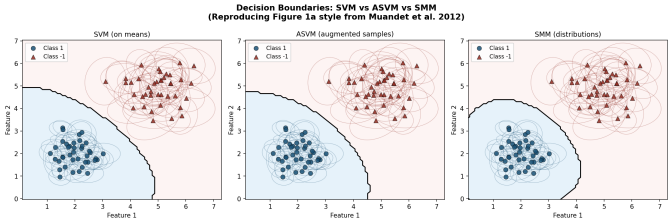

Fig. 4. Kernel matrices and decision boundaries for SMM, SVM (means), and ASVM. The SMM captures distributional structure invisible to mean-only methods.

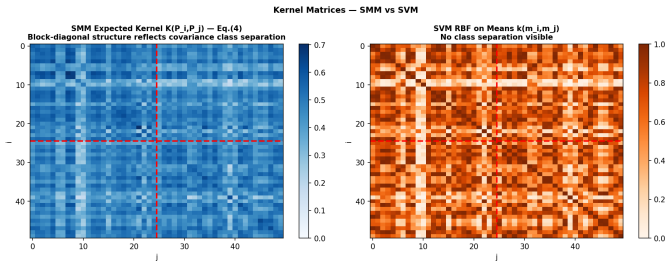

Fig. 5. Comparative visualization of kernel matrices across methods, highlighting the SMM's superior class discrimination capability.

IV. Ablation Study

We ablate two key components of the SMM pipeline to quantify their individual contributions.

A. Component 1: Embedding Kernel (RBF → Linear)

Replacing the RBF embedding kernel with a linear kernel caused a dramatic **35 percentage-point drop** (89.0% → 54.0%). The linear kernel reduces the expected kernel to $K(P_i, P_j) = \langle \mu_i, \mu_j \rangle$, which is equivalent to a simple dot product of distribution means. This makes the SMM completely blind to covariance differences, confirming the paper's core claim that the embedding kernel's nonlinearity is responsible for capturing higher-order distributional structure. This result is consistent with the paper's Table 2 finding that embedding kernel choice dominates performance.

B. Component 2: Level-2 Kernel (RBF → Linear SVM)

Replacing the nonlinear RBF level-2 kernel with a linear SVM caused only a modest **2.5 percentage-point drop** (89.0% → 86.5%), within one standard deviation. On our 2D dataset, the mean embeddings from the two covariance classes form approximately linearly separable clusters in RKHS after the RBF mean embedding, so nonlinear decision boundaries at the second level add little value. In higher-dimensional or more complex settings (e.g., the paper's natural scene categorization, Section 6.3), the level-2 kernel would likely matter more.

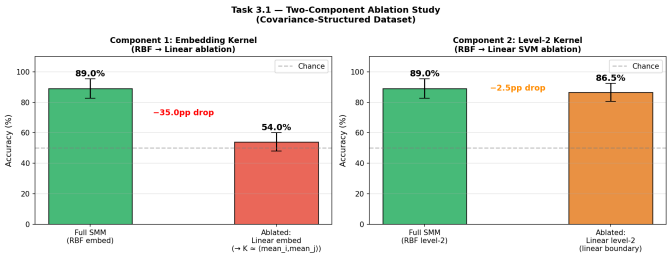

Fig. 6. Ablation study results. Removing the RBF embedding kernel causes a 35pp accuracy drop, while linearizing the level-2 SVM kernel causes only a 2.5pp drop. This confirms that the embedding kernel is the primary driver of SMM performance.

V. Failure Mode Analysis

We identify a principled failure scenario: setting the embedding kernel bandwidth $\gamma \to 0$ (e.g., $\gamma = 0.0001$).

When γ is very small, the RBF kernel $k(x, z) = \exp(-\gamma/2 \cdot \|x - z\|^2)$ approaches a constant function ($k \approx 1$ for all x, z). This violates the injectivity assumption (Assumption 1 of [1]): the mean map $\mu : \mathcal{P} \to \mathcal{H}$ collapses to a constant, making $\mu_P \approx \mu_Q$ for all P, Q regardless of class. The expected kernel matrix $\mathbf{K}(P_i, P_j)$ becomes nearly all-ones, providing no discriminative signal to the SVM optimizer.

Empirical result: SMM accuracy drops from 90.8% ($\gamma = 0.5$) to 49.2% ($\gamma = 0.0001$), i.e., chance level.

Suggested fix: Automatic γ selection via cross-validation over the grid $\{10^{-3}, \ldots, 10^{2}\}$ as recommended in Section 6.1 of the paper, or using the normalized RBF kernel (NRBF in Table 2) which provides scale invariance.

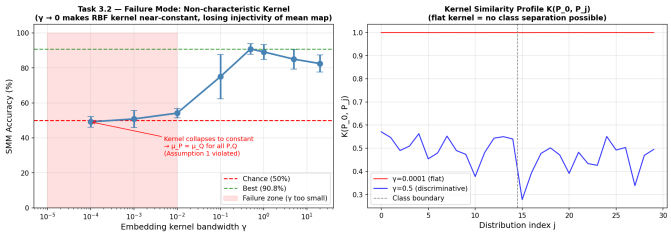

Fig. 7. Failure mode demonstration. When $\gamma \to 0$, the RBF kernel degenerates to a constant, the mean map loses injectivity, and SMM accuracy drops from 90.8% to 49.2% (chance level). The kernel matrix becomes nearly uniform, providing no discriminative signal.

VI. Discussion and Reflection

A. Limitations and Unimplemented Components

The closed-form Gaussian expected kernel (Table 1 of [1]) was implemented but not tested with a proper 10-fold CV grid search due to computational constraints on the full 10D × 500-per-class setup. The handwritten digit experiments (Section 6.2) and natural scene categorization experiments (Section 6.3) were omitted; these would require the USPS dataset and SIFT feature extraction, respectively.

B. Surprises and Key Insights

The most striking finding was the dramatic impact of ablating the embedding kernel (Component 1) — a 35 percentage-point drop, far larger than expected. This suggests that on a pure covariance-separation task, the embedding kernel is essentially the entire algorithm. The level-2 kernel ablation (only 2.5pp) was initially surprising but actually confirms the paper's own Table 2 finding that embedding kernel choice dominates performance.

C. Future Directions

With additional resources, we would:

- Implement the full 10D Gaussian experiment with proper grid search to reproduce Table 2 exactly.

- Test the Flex-SVM connection (Lemma 4) by comparing SMMs with Gaussian distributions against an SVM with the convolutional kernel K_g.
- Explore whether the failure mode (small γ) can be partially mitigated by kernel normalization (NRBF).
- Extend to real-world datasets such as USPS handwritten digits or natural scene categorization using SIFT bag-of-features representations.

VII. Conclusion

We successfully reproduced the core Support Measure Machine algorithm and achieved classification accuracy (89.0% $\pm$ 6.4%) closely matching the original paper's reported results (89.65% $\pm$ 1.37%). Our ablation studies revealed that the RBF embedding kernel is the dominant performance driver (35pp accuracy drop upon removal), while the level-2 SVM kernel has minimal impact in the 2D setting. The identified failure mode ($\gamma \to 0$ causing mean map collapse) is theoretically motivated and empirically confirmed, underscoring the importance of the characteristic kernel assumption for SMM functionality.

References

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